



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT		
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Abstract

Traffic congestion is a major challenge in urban areas, leading to increased travel time, fuel consumption, and environmental pollution. Conventional traffic signal systems operate on fixed time intervals and are unable to adapt to real-time traffic conditions, resulting in inefficient traffic management. This project proposes an AI-Powered Smart Traffic Signal System using Reinforcement Learning that dynamically adjusts traffic signal timings based on live traffic density.

The system utilizes computer vision techniques with OpenCV and YOLO to detect and count vehicles from camera feeds. The collected traffic data is processed by a Reinforcement Learning (RL) model, which learns optimal signal timing strategies by interacting with simulated traffic environments and maximizing traffic flow efficiency. Unlike traditional systems, the proposed solution continuously adapts to changing traffic conditions, reducing vehicle waiting time, congestion, and unnecessary fuel consumption. A user-friendly web dashboard is developed using React.js and FastAPI to display real-time traffic statistics, signal status, and system performance. The project also incorporates modern software engineering practices, including modular architecture, RESTful APIs, Git-based version control, and containerized deployment using Docker. Additionally, an AI assistant powered by a Large Language Model (LLM) can provide traffic insights, explain signal decisions, and generate analytical reports.

The proposed system demonstrates how adaptive software engineering and artificial intelligence can be combined to create an intelligent, scalable, and efficient traffic management solution. It offers a practical approach for developing smart city infrastructure while showcasing the application of machine learning, computer vision, and modern web technologies in solving real-world transportation challenges.

Signature of the Guide

HOD